Causal Inference Workshop

Week 6 - TWFE
Application and Implementation

Causal Inference Workshop

February 27, 2024

Workshop outline

A. Causal inference fundamentals
   - Modeling assumptions matter too
   - Conceptual framework (potential outcomes framework)
B. Design stage: common identification strategies
   - IV + RDD [coding]
   - Event Studies, DiD, DiDiD, New TWFE Lit [coding]
   - Synthetic Control / Synthetic DiD [coding]
C. Analysis stage: strengthening inferences
   - Limitations of identification strategies, pre-estimation steps
   - Estimation [controls] and post-estimation steps [supporting assumptions]
D. Other topics in causal inference and sustainable development
   - Inference (randomization inference, bootstrapping)
   - Weather data regressions, other common/fun SDev topics [coding]
   - Remote sensing data, other common/fun SDev topics

Causal inference roadmap

- *Potential outcomes* [framework]
  - Causal effect is the difference between two potential outcomes
  - We can't observe this difference, but can see differences in average observed outcomes
  - If **(conditional) independence assumption** holds, can estimate unbiased ATT
- *Identification* [application/implementation] [last week, and today, ... and next week!]
  - In most empirical settings, IA and CIA do not hold, which is why we need an **identification strategy**
  - Want to eliminate selection bias (identification problem)
- *Estimation* [application/implementation]
  - (Usually) use linear regression model
  - $\hat{\beta}_{OLS}$ unbiased estimator for ATT if e is uncorrelated with treatment (regression problem)

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Two-Way Fixed Effects (TWFE)

Two-way fixed effects estimators with heterogeneous treatment effects C De Chaisemartin, X d'Haultfoeuille American economic review 110 (9), 2964-2996	6586	2020
Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey C De Chaisemartin, X d'Haultfoeuille The econometrics journal 26 (3), C1-C30	1337	2023
Difference-in-differences with multiple time periods B Callaway, PHC Sant'Anna Journal of econometrics 225 (2), 200-230	10617	2021
Difference-in-differences with a continuous treatment B Callaway, A Goodman-Bacon, PHC Sant'Anna National Bureau of Economic Research	1295	2024
Difference-in-differences with variation in treatment timing A Goodman-Bacon Journal of econometrics 225 (2), 254-277	10349	2021

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Two-Way Fixed Effects (TWFE)

- Review of TWFE problem
- Suggestions for implementation
- Coding TWFE estimators

TWFE problem overview

- Canonical 2x2 DiD (2 periods and 2 groups, treated groups experience treatment in period 2) **great** – easy to understand what is estimated and assumptions needed

- However, most actual research these days is **not** using 2x2 DiD, but rather TWFE:

$$y_{it} = \lambda_i + \delta_t + \beta_{TWFE} D_{it} + e_{it}$$

- Problems begin because we do not (or did not) actually understand how this estimator is comparing mean outcomes across groups

- Big picture takeaway: TWFE regressions estimate *weighted sums* of the ATE in each group and period, with weights that sum to one but may make TWFE estimator biased (and *may* even reverse the sign)

TWFE problem overview

- Big picture takeaway: TWFE regressions estimate *weighted sums* of the ATE in each group and period, with weights that sum to one but may make TWFE estimator biased (and *may* even reverse the sign)

$$y_{it} = \lambda_{g[i]} + \delta_t + \beta_{TWFE} D_{g[i]t} + e_{g[i]t}$$

$\rightarrow$ $\hat{\beta}_{TWFE}$ is a specific weighted sum of the ATE in each treated (g,t) cell (Chaisemartin and D'Haultfœuille 2020) $ATE_{gt} = \frac{1}{N_{gt}} \sum_{i=1}^{N_{gt}} [Y_{i,gt}(1) - Y_{i,gt}(0)]$.

$$\beta_{ATT} = \mathbb{E}\left[\sum_{(gt):D_{gt}=1} \frac{N_{gt}}{N_1} ATE_{gt} \right], \qquad \mathbb{E}[\hat{\beta}_{TWFE}] = \mathbb{E}\left[\sum_{(gt):D_{gt}=1} \frac{N_{gt}}{N_1} w_{gt} ATE_{gt} \right]$$

$\rightarrow$ weights may vary and can even be **negative!** (Chaisemartin and D'Haultfœuille 2020)
negative weights huge potential issue, as it may be $\hat{\beta}_{TWFE} < 0$ even if all $ATE_{gt} > 0$ (e.g., $1.5 \times 1 - 0.5 \times 4$)...)

More on TWFE weights

- TWFE regressions estimate *weighted sums* of the ATE in each group and period

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- Weights are proportional to and have the same sign as

$$w_{g,t} \propto (D_{g,t} - D_{g,.} - D_{.,t} + D_{.,.})$$

More on TWFE weights

- Weights are proportional to and have the same sign as

$$w_{g,t} \propto (D_{g,t} - D_{g,.} - D_{.,t} + D_{.,.})$$

  - $D_{g,t}$: treatment in group g at period t
  - $D_{g,.}$: average treatment of group g across periods
  - $D_{.,t}$: average treatment at period t across groups
  - $D_{.,.}$: average treatment across groups and periods

- $\hat{\beta}_{TWFE}$ is unbiased for ATT
  - if ATE_{gt} is exactly the same for every group and every period, or
  - if weights are uncorrelated with the conditional ATEs in the treated (g,t) cells. But in most cases, no-correlation condition is implausible.
    - For example, groups treated most are also those with the lowest weight. But those groups could also be those with the largest treatment effects.

- Note specifically that $\hat{\beta}_{TWFE}$ downweights TEs of groups with highest average treatment and time periods with highest average treatment

An Example for Chaisemartin and D'Haultfœuille 2020 Weights

Two groups of equal size:

Table: $D_{g,t}$

	t=1	t=2	t=3
g=1	0	1	1
g=2	0	0	1

$$w_{g,t} \propto (D_{g,t} - D_{g,.} - D_{.,t} + D_{.,.})$$

$$g = 1, t = 2: \quad 1 - 2/3 - 1/2 + 1/2 = 1/3,$$

$$g = 1, t = 3: \quad 1 - 2/3 - 1 + 1/2 = -1/6,$$

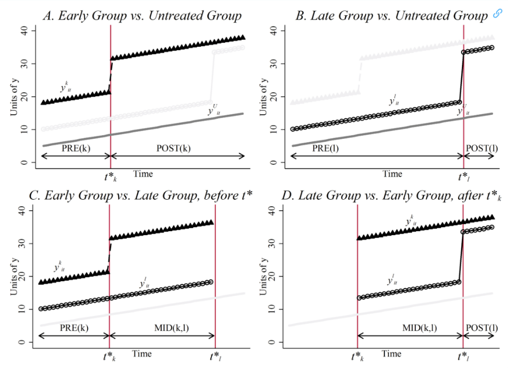
$$g = 2, t = 3: \quad 1 - 1/3 - 1 + 1/2 = 1/6.$$

Normalizing the weights so that their sum equals one. This leads to

$$\mathbb{E}[\hat{\beta}_{TWFE}] = 1/2\mathbb{E}[ATE_{2,3}] + \mathbb{E}[ATE_{1,2}] - 1/2\mathbb{E}[ATE_{1,3}].$$

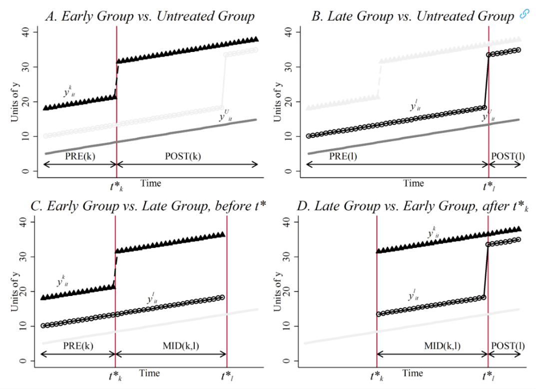
Goodman-Bacon 2021 Decomposition

- A different type of decomposition (same issue but different angle): all the DD comparisons get positive weights but problem originates from comparing newly treated to previously treated groups (Goodman-Bacon 2021)



- $\mathbb{E}[\hat{\beta}_{TWFE}] \neq ATT$.

Goodman-Bacon 2021 Decomposition

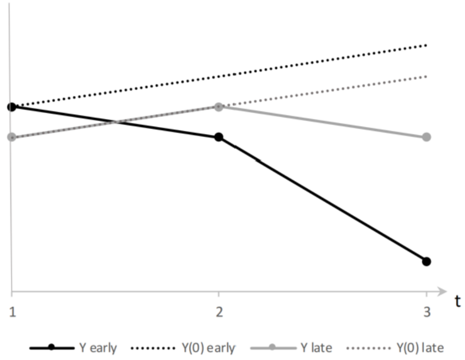


$$\hat{\beta}_{TWFE} = \underbrace{\sum_{k \neq U} s_{kU} \hat{\beta}^{2x2}_{kU}}_{\text{ever-treated vs never treated}} + \underbrace{\sum_{k \neq U} \sum_{\ell > k} s^{k}_{k\ell} \hat{\beta}^{2x2,k}_{k\ell}}_{\text{already treated vs not-yet-treated}} + \underbrace{\sum_{k \neq U} \sum_{\ell > k} s^{\ell}_{k\ell} \hat{\beta}_{k\ell^{2x2,\ell}}}_{\text{later treated vs early treated}} .$$

- $\mathbb{E}[\hat{\beta}_{TWFE}] \neq ATT$.
- The extreme case is that $\hat{\beta}_{k\ell^{2x2,\ell}}$ is so negative, even though the treatment effect is positive, that the sign of the estimator is reversed.

An Example of "Forbidden" Comparison Being of Opposite Sign (from Chaisemartin and D'Haultfœuille 2023)

Figure 6.1: A tale of two patients who had an infection.



	t=1	t=2	t=3
g=1	0	1	1
g=2	0	0	1

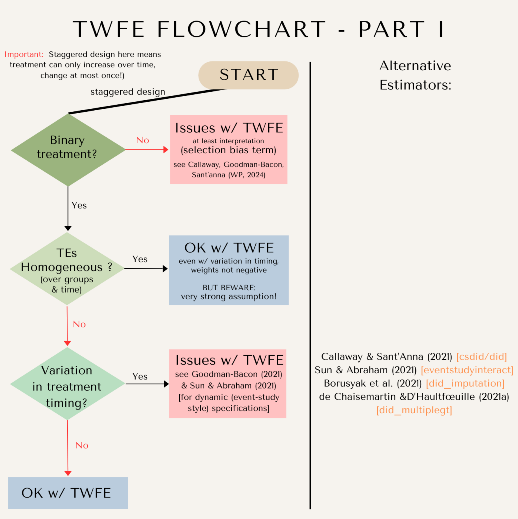
- 2X2 DD of late adopters (treated) and early adopters (control) from period 2 to 3.
  $[Y_{2,3}(1) - Y_{2,2}(0)] - [Y_{1,3}(1) - Y_{1,2}(1)]$
- Although the antibiotic suppresses fever (lower temperature) for both patients, the above DD is positive because the early adopter experienced an enhanced effect during her second period of taking the medicine.
- Varying treatment effect over time is causing the problem (problem exists even if both groups follow the exact dynamic treatment effect trajectory).

TWFE and when it's a problem

1. You should worry when **treatment effects are heterogeneous** (not constant across groups or over time) – this is *most* studies!
2. You should also understand stronger assumptions with **continuous** treatments
3. Definitional point: often people refer to "staggered" designs (technically means treatment can only increase, BUT people usually refer to binary treatments that can only increase, but can increase at any time as "staggered")

- Suggested reading order:
  - Chaisemartin and D'Haultfœuille 2023 review: *Two-Way Fixed Effects and Differences-in-Differences with Heterogeneous Treatment Effects: A Survey*
  - Chaisemartin and D'Haultfœuille 2020 (AER, overview of problem & alternate estimator)
  - Goodman-Bacon (2021) helpful for understanding intuition behind negative weights and "forbidden comparisons" (using early-treated group as control for late-treated group)
    → then decide what else you need to learn about for your specific set-up!

TWFE and when it's a problem



Overview of alternative estimators

For binary and staggered designs (with heterogeneous treatment effects):

- Methods that calculate individual treatment effects and then aggregate to single ATT
  - Callaway and Sant'Anna (2021) [csdid]: never-treated units or not-yet-treated units as controls. Use the doubly robust method to take advantage of observed pre-treatment covariates.
  - Sun and Abraham (2021) [eventstudyinteract]: never-treated or last treated as controls.
  - de Chaisemartin and D'Haultfoeuille (2020) [did_multiplegt]

- Borusyak et al. (2021) [didimputation]: imputation based, estimate missing counterfactuals in multi-step process (extrapolate potential outcomes for the treated units, by using data for those observations that were not treated yet)

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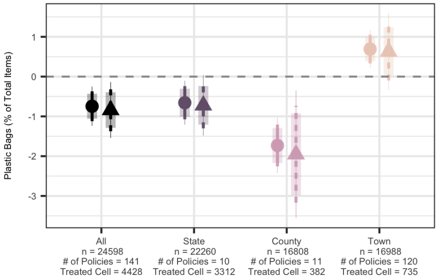
- Example (Papp and Oremus 2025):
  - Effect of plastic bag bans and taxes on plastic in shoreline cleanups across the US
  - binary treatment, heterogeneous treatment effects, variation in treatment timing
  - (unbalanced panel, so implement Borusyak et al. as the main estimator, but show robustness to de Chaisemartin and d'Haultfoeuille in appendix)



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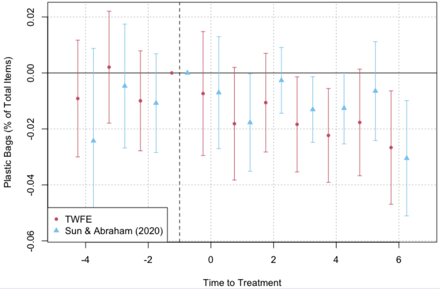

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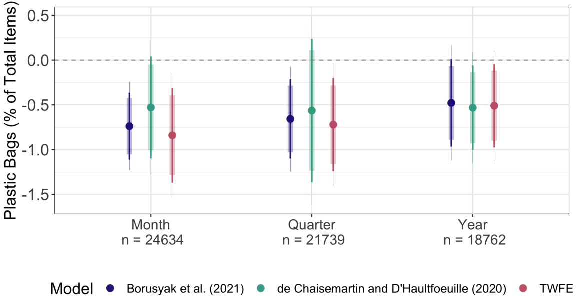

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Coding new TWFE estimators, data

- Stevenson and Wolfers (2006), *QJE*
  - Explores the impacts of divorce law reform in the United States
  - Exploits the variation in the timing of unilateral divorce laws across states
  - Impact on female suicide rates, domestic violence, and murders of women by partners
    → decrease in all of these
- Cheng and Hoekstra (2013), *Journal of Human Resources*
  - Explores the impacts of "Castle Doctrine" ("stand your ground" laws, which expand legal justification for using lethal force in self-defense)
  - Exploits variation in the timing of these laws in 20+ states
  - Impact on homicide and violent crime
    → increase in reported murders
- → binary treatment, heterogeneous treatment effects, variation in treatment timing

Coding new TWFE estimators, summary

Use: `01_twfe.R`

- Load data (Stevenson and Wolfers by default)
- Old school TWFE and dynamic event-study version
- Bacon decomposition (earlier vs. later treated, etc.)
- Implement and compare new TWFE estimators
  - Sun and Abraham (2021)
  - Callaway and Sant'anna (2021)
  - Borusyak et al. (2021)
  - de Chaisemartin and D'Haultfoeuille (2020)

Questions? Comments?

Thank you!

References I

Heavily based on Claire Palandri's 2022 version and Anna Papp's 2024 version of the Causal Inference Workshop.

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Stevenson, Betsey, and Justin Wolfers. 2006. "Bargaining in the Shadow of the Law: Divorce Laws and Family Distress*." *The Quarterly Journal of Economics* 121 (1): 267–288. ISSN: 0033-5533. https://doi.org/10.1093/qje/121.1.267. eprint: https://academic.oup.com/qje/article-pdf/121/1/267/5230654/121-1-267.pdf. https://doi.org/10.1093/qje/121.1.267.